

Formula 1 Forecast: Build Your Grid

Milestone #: 1

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Group Number: 61

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Description

The domain of our project is in Sports, Fan Engagement and Community Building. Implementing a Formula 1 prediction platform providing Users with the chance to predict outcomes of race weekends throughout the formula 1 season, earning points based on accuracy. Additionally, the application will allow multiple users to compete together by comparing their scores.

Language

SQL and JavaScript

Changes

No changes suggested

An explicit acknowledgment about your use of AI tools in this assignment. Specifically, We are looking for a clear yes/no about whether you have used one or more AI tools. If yes, we want to know which tool(s) you have used and we want a PDF of the full Conversation.

No generative AI was used in this proposal.

Relational Schema:

Category_Nameld(Name, Id)

Category_NameDesc(Name, Description)

- Name → References Category_Nameld(Name)

Category_NameSession(Name, SessionType)

- Name → References Category_Nameld(Name)

Category_IdDesc(Id, Description)

- Id → References Category_Nameld(Id)

Prediction(PredictionId, PredictionValue, DateAndTimeFiled)

PCHas(CategoryId, PredictionId)

- CategoryId → References Category_Nameld(Id)
- PredictionId → References Prediction(PredictionId)

PredictionFor(PredictionValue, DateFiled, PredictionId, Season, TrackName)

- PredictionId → References Prediction(PredictionId)

PredictionMakes(PredictionValue, DateFiled, PredictionId, DateOfFirstPrediction)

- PredictionId → References Prediction(PredictionId)

PSHas(Ranking, Acc, PredictionId)

- PredictionId → References Prediction(PredictionId)

Scores(Acc, Ranking, Amount, Deductions)

Session(Season, TrackName, Date)

Driver_Core(DriverId, Name, AccumulatedPoint)

Driver_NameEntry(Name, DateOfEntryToF1)

- Name → References Driver_Core(Name)

Team_Core(TeamId, Points, Name)

Team_PointsNameEntry(Points, Name, DateOfEntryToF1)

- Points → References Team_Core(Points)
- Name → References Team_Core(Name)

Team_IdEntry(TeamId, DateOfEntryToF1)

- TeamId → References Team_Core(TeamId)

Results(Position, TotalTime, DriverOfTheDay, Pitstops)

User(Id, DateOfFirstPrediction, Name, Streak)

FFHas(Friend1, Friend2)

- Friend1 → References User(Id)
- Friend2 → References User(Id)

Race(RaceId, TrackName, Season)

- TrackName → References Session(TrackName)
- Season → References Session(Season)

Sprint(SprintId, TrackName, Season)

- TrackName → References Session(TrackName)
- Season → References Session(Season)

Qualifying(QualifyingRound, TrackName, Season)

- **TrackName** → References Session(TrackName)
- **Season** → References Session(Season)

Practice(PracticeRound, TrackName, Season)

- **TrackName** → References Session(TrackName)
- **Season** → References Session(Season)

Normalization to 3nf:

Category:

Id -> Name, Description, SessionType

Name -> Description

Decomposition to 3nf:

Id -> Name

Id -> Description

Id -> SessionType

Name -> Description

Candidate Keys = {Id}

Name -> Description violates bcnf

R1(Name, Description)

R2(Name, Id, SessionType)

R1 is in bcnf, r2 is in bcnf

Scores:

Acc, Ranking -> Amount, Deductions

Prediction:

PredictionId -> PredictionValue, DateAndTimeFiled

Session:

Season, TrackName -> Date

Date -> Season

Decomposition to 3nf using synthesis:

Season from Date-> Season is present within the primary key. This means Session is already in 3nf

Driver:

DriverId -> AccumulatedPoint, DateOfEntryToF1, Name

Name -> DateOfEntryToF1

Decomposition to 3nf using synthesis

Candidate Key: {DriverId}

Name -> DateOfEntryToF1 violates 3nf

Decompose to R1(Name, DateOfEntryToF1), R2(Name, DriverId, AccumulatedPoint)

R2 is in 3nf since only valid FD's represent the key: DriverId -> AccumulatedPoints, DriverId -> Name

Team:

TeamId -> Points, Name, DateOfEntryToF1

Name -> DateOfEntryToF1

Points, Name -> DateOfEntryToF1

DateOfEntryToF1, Points -> Name

Decomposition to 3nf using synthesis

TeamId -> Points

TeamId -> Name

TeamId -> DateOfEntryToF1

Points, Name -> DateOfEntryToF1

DateOfEntryToF1, Points -> Name

Candidate Keys = {TeamId}

Decompose on Points, Name -> DateOfEntryToF1

R1 (Points, Name, DateOfEntryToF1), R2(TeamId, Points, Name)

Add back missing FD's

R3(TeamId, DateOfEntryToF1)

Results:

Position -> DriverOfTheDay, Pitstops, TotalTime

Position -> Total Time

TotalTime -> Position

Decomposition to 3nf using synthesis

Position -> DriverOfTheDay

Position -> Pitstops

Position -> TotalTime

Position -> TotalTime

TotalTime -> Position

Candidate Keys = {TotalTime, Position}

In bcnf and 3nf already

User:

Id, DateOfFirstPrediction -> Name, Streak

SQL DDL:

```
create table CategoryDescription
(
    category_name      varchar,
    category_description varchar not null,
    primary key category_name,
    foreign key (category_name) references Category (category_name)
);
```

```
create table CategorySessionType
(
    category_name varchar,
    sessionType   varchar,
    primary key category_name,
    foreign key (category_name) references Category (category_name)
);
```

```
create table Category
(
    category_name varchar,
    category_id   varchar,
    primary key category_id
);
```

```
create table CategoryDetails
(
    category_id      varchar,
    category_description varchar,
    primary key category_id,
    foreign key (category_id) references Category (category_id)
);
```

```
create table Prediction
(
    predictionId      varchar,
    predictionValue   number not null,
    dateAndTimeFiled date   not null,
    primary key predictionId
);
```

```
create table PredictionCategory
(
    categoryId   varchar,
    predictionId varchar,
    primary key (categoryId, predictionId),
    foreign key (categoryId) references Category (category_id),
);
```

```

        foreign key (predictionId) references Prediction (predictionId)
    );

create table PredictionForSession
(
    predictionValue number,
    dateFiled      date,
    predictionId   varchar,
    season         varchar,
    trackName      varchar,
    primary key (predictionId, season, trackName),
    foreign key (predictionId) references Prediction (predictionId)
);

create table PredictionValues
(
    predictionValue      number,
    dateFiled            date,
    predictionId          varchar,
    dateOfFirstPrediction date,
    primary key predictionId,
    foreign key (predictionId) references Prediction (predictionId)
);

create table PredictionRanking
(
    ranking      number,
    acc          varchar,
    predictionId varchar,
    primary key (ranking, acc, predictionId),
    foreign key (predictionId) references Prediction (predictionId)
);

create table AccountScore
(
    acc          varchar,
    ranking      number,
    amount       number,
    deductions   number,
    primary key (acc, ranking)
);

create table PracticeSession
(
    practiceRound number,

```

```
trackName      varchar,
season         varchar,
primary key (practiceRound),
foreign key (trackName, season) references Session (trackName, season)
);

create table QualifyingSession
(
    qualifyingRound number,
    trackName      varchar not null,
    season         varchar not null,
    primary key (qualifyingRound),
    foreign key (trackName, season) references Session (trackName, season)
);

create table SprintSession
(
    sprintId  varchar,
    trackName varchar not null,
    season    varchar not null,
    primary key (sprintId),
    foreign key (trackName, season) references Session (trackName, season)
);

create table results
(
    position      varchar,
    totalTime     varchar,
    driver_of_the_day varchar,
    num_pitstops  varchar,
    primary key (position, totalTime)
);
```